

Problem

Design a problem to help other students better understand solving integrodifferential equations with a periodic input, using Laplace transforms.

Step-by-step solution

Step 1 of 1

Although there is no correct way to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Solve the following integrodifferential equation using the Laplace transform method:

$$\frac{dy(t)}{dt} + 9 \int_0^t y(t) dt = \cos 2t, \quad y(0) = 1$$

Solution

Take the Laplace transform of each term.

$$[sY(s) - y(0)] + \frac{9}{s} Y(s) = \frac{s}{s^2 + 4}$$

$$\left(\frac{s^2 + 9}{s} \right) Y(s) = 1 + \frac{s}{s^2 + 4} = \frac{s^2 + s + 4}{s^2 + 4}$$

$$Y(s) = \frac{s^3 + s^2 + 4s}{(s^2 + 4)(s^2 + 9)} = \frac{As + B}{s^2 + 4} + \frac{Cs + D}{s^2 + 9}$$

$$s^3 + s^2 + 4s = A(s^3 + 9s) + B(s^2 + 9) + C(s^3 + 4s) + D(s^2 + 4)$$

Equating coefficients :

$$s^0: \quad 0 = 9B + 4D$$

$$s^1: \quad 4 = 9A + 4C$$

$$s^2: \quad 1 = B + D$$

$$s^3: \quad 1 = A + C$$

Solving these equations gives

$$A = 0, \quad B = -4/5, \quad C = 1, \quad D = 9/5$$

$$Y(s) = \frac{-4/5}{s^2 + 4} + \frac{s + 9/5}{s^2 + 9} = \frac{-4/5}{s^2 + 4} + \frac{s}{s^2 + 9} + \frac{9/5}{s^2 + 9}$$

$$y(t) = [-0.4 \sin(2t) + \cos(3t) + 0.6 \sin(3t)] u(t)$$